Huawei Access Node QoS Congestion Event Resolution SOP

Node: WFD-ACC-18 | **Vendor:** Huawei | **Severity:** Critical | **Alarm Code:** ALARM_3387

Description

This Standard Operating Procedure (SOP) addresses a critical QoS congestion event on Huawei Access Node WFD-ACC-18 located at Whitefield POP, which manifests as ALARM_3387. This failure domain is attributed to microbursts and interface tail drops resulting from QoS queue starvation. The underlying cause is rooted in the oversubscription of traffic flows to the Access Node's QoS queues, compounded by inadequate policing and shaping mechanisms, which ultimately lead to the collapse of the queue management system. This collapse causes packets to be discarded at the interface tail, inducing microbursts and triggering the ALARM_3387 alarm.

Pre-requisite Checks

- pre-requisite: Confirm the physical and environmental status of the Whitefield POP Site, including power, cooling, and any environmental factors that could impact the Access Node's operation.

- pre-requisite: Verify the redundant link status for the Access Node to ensure no Single Point of Failure (SPOF) exists.

- pre-requisite: Ensure access to the Huawei Command-Line Interface (CLI) for the Access Node WFD-ACC-18.

Potential Root Causes

Cause	Probability
Inadequate policing and shaping of traffic flows on the Access Node	High
Oversubscription of traffic flows to QoS queues on the Access Node	Medium
Hardware or software defects in the Access Node's QoS management system	Low

Diagnostic Commands & Expected Outputs

```
Command: display queue statistics, Expected Output: Queue statistics including packet drops, queue length, and average queue length, Description: This command isolates the queue statistics, providing insights into the QoS queue utilization and packet drops on the Access Node.

Command: display interface counters, Expected Output: Interface counters including packet drops, errors, and discards, Description: This command isolates the interface counters, providing insights into the interface utilization and packet drops on the Access Node.
```

Resolution Steps

Step 1: Step: Verify and adjust policing and shaping parameters, Command: conf t policy-map class class-default police 1000000 service-policy input, Description: Adjust the policing and shaping parameters to prevent oversubscription of traffic flows to QoS queues, thus preventing QoS queue starvation.

Step 2: Step: Verify and adjust QoS queue configuration, Command: conf t queue-set queue-set 1 queue 1 default queue 2 default, Description: Adjust the QoS queue configuration to ensure proper policing and shaping of traffic flows, thus preventing QoS queue starvation.

Post-Check Verification

```
Command: display queue statistics, Expected Output: Queue statistics indicating no packet drops or queue length exceeding threshold, Description: This command verifies that the QoS queue utilization and packet drops on the Access Node have been resolved.

Command: display interface counters, Expected Output: Interface counters indicating no packet drops or errors, Description: This command verifies that the interface utilization and packet drops on the Access Node have been resolved.
```

Roll-back Procedures

- Step: Restore original policing and shaping parameters, Command: conf t policy-map class class-default no police 1000000 no service-policy input, Description: Restore the original policing and shaping parameters if the fix fails, thus reverting to the previous configuration.

- Step: Restore original QoS queue configuration, Command: conf t queue-set queue-set 1 no queue 1 default no queue 2 default, Description: Restore the original QoS queue configuration if the fix fails, thus reverting to the previous configuration.